Integration by Parts

$$\int f(x)\, g'(x)\, dx = f(x)\, g(x) - \int f'(x)\, g(x)\, d$$

$$\int x \cdot \underbrace{\boxed{\cos x}}_{\sin t'}\, dx = x \cdot \sin x - \int x' \sin x$$

$$= x \cdot \sin x - \int \sin x\, dx = x \cdot \sin x + \cos x + C$$

$$\int 2^x \cdot x\, dx = \int \left(\frac{2^x}{\ln 2}\right)' \cdot x\, dx =$$

$$\frac{2^x}{\ln 2} \cdot x - \int x' \frac{2^x}{\ln 2} = \frac{2^x}{\ln 2} - \frac{1}{\ln 2} \cdot \int 2^x$$

$$= \frac{2^x}{\ln 2} - \frac{2^x}{(\ln 2)^2}$$

3) $x^2 e^x = x^2 \cdot (e^x)' = x^2 e^x - \int (x^2)' e^x$

$$= \underbrace{x^2 e^x - 2}\int x \cdot e^x\, dx =$$

$$x \cdot (e^x)' = x \cdot e^x - \int x' e^x = x \cdot e^x - e^x$$

$$x^2 e^x - 2(x \cdot e^x - e^x) + C$$

4) $\int \ln x \, dx = \int 1 \cdot \ln x \, dx =$

$$\int x' \ln x \, dx = x \cdot \ln x - \int x \cdot \ln x'$$

$$x \cdot \ln x - \int x \cdot \frac{1}{x} = x \cdot \ln x - x$$

U-sub

1. $\int x \cdot \cos(x^2) \, dx$

let $u = x^2 \, dx$

$du = 2x \qquad x = \frac{du}{2}$

$$\int \cos(u) \frac{du}{2} = \frac{1}{2} \int \cos(u) \, du$$

$$\frac{1}{2} \sin(u) + C = \frac{1}{2} \sin(x^2) + C$$

2) $\int x^2 \sqrt[3]{4x^3+3}\, dx$

let $u = 4x^3+3$ dx

$du = 12x^2$

$x^2 = \frac{du}{12}$

$\frac{1}{12} \int \sqrt[3]{12x^2}\, du = \frac{1}{12} \int 12u^{\frac{2}{3}}\, du$

$\frac{1}{12} \cdot \frac{12u^{\frac{4}{3}}}{\frac{4}{3}} + C$

3) $\int \sin(10x)\, dx$

let $u = 10x$

$du = 10\, dx$

$dx = \frac{du}{10}$

$\frac{1}{10} \int \sin(u)\, du = \frac{1}{10} -\cos(u) + C$

$$\frac{1}{10} \cdot -\cos(10x) + C$$

$$\int_1^8 \frac{2+x}{\sqrt[3]{x}}\, dx \qquad \sqrt[3]{x} = x^{\frac{1}{3}}$$

$$\frac{2}{x^{\frac{1}{3}}} + \frac{x}{x^{\frac{1}{3}}}$$

$$2x^{-\frac{1}{3}} + x^{\frac{2}{3}} = \frac{2x^{\frac{2}{3}}}{\frac{2}{3}} + \frac{x^{\frac{5}{3}}}{\frac{5}{3}}$$

$$\int \sec^4 x \cdot \tan^4 x$$

$$\int \sin^m x \cos^n x \, dx$$

$n = 2k+1$ $\qquad$ $\cos^2 x = 1 - \sin^2 x$

$$\int \sin^m x \cdot \cos^n x = \int \sin^m x \, (\cos^2 x)^k \cos x \, dx$$

$$\int \sin^m x \, (1 - \sin^2)^k \cos x \, dx$$

let $u = \sin x$

$du = \cos x \, dx$

$m = 2k+1$ $\qquad$ odd

$$\int (1 - \cos^2 x)^k \cos^n x \sin x \, dx$$

let $u = \cos x$

$$\frac{x}{x^2+x-2} = (x+2)(x-1)$$

$$\frac{A}{x+2} + \frac{B}{x-1}$$

A, B

$$A(x-1) + B(x+2)$$

$$Ax - A + Bx + 2B$$

$$x(A+B) + (2B-A)$$

$1 \cdot x = 0 \quad (A+B)x + (2B-A)$

$$\begin{cases} A+B = 1 \\ 2B-A = 0 \end{cases} \qquad A = \frac{2}{3} \qquad B = \frac{1}{3}$$

$$\frac{x}{x^2-x-2} = \frac{2/3}{x+2} + \frac{1/3}{x-1}$$

$$\int \sec^4 x \cdot \tan^4 x \quad dx$$

$$\tan^4 x\,(1+\tan^2 x)\cdot \sec^2 x$$

let $u = \tan x$

$$u^4(1+u^2)\,du$$

$$u^4 + u^6 \qquad \frac{u^5}{5} + \frac{u^7}{7}$$

c)

$$\frac{x+1}{2x^3+x^2} \qquad \frac{A}{x} + \frac{B}{x^2} + \frac{C}{2x+1}$$

$$x^2(2x+1)$$

$$y = 3-x \qquad xy = 2 \qquad y = \frac{2}{x}$$

$$y = 3-x$$

$$x(3-x) = 2$$

$$3x - x^2 = 2$$

$$x^2 - 3x + 2 = 0$$

$$(x-1)(x-2)$$

$$x = 1 \qquad x = 2$$

$$y = \frac{2}{x} \qquad y = 3 - x$$

$$\int_1^2 \left[(3-x) - \frac{2}{x} \right] dx$$

$$x - 1 = y^2 \qquad x = 5$$

$$x = y^2 + 1$$

$$5 = y^2 + 1$$

$$x = 5$$
$$x = y^2 + 1$$

$$y = \pm 2$$

$$\int_{-2}^{2} 5 - (y^2 + 1)\, dy$$

$$y = 4 - x^2$$

$$x^2 = 4 \qquad x = \pm 2$$

$$y = -3$$

$$y = 4 - x^2$$

$$x^2 = 4 - y$$

$$x = \pm\sqrt{4-y}$$

$$2\sqrt{4-y}$$

$$y = x^3 \qquad x = y^2$$

$$x = x^3$$

$$x^3 - x = 0$$

$$x(x^2-1) = 0$$

$$x = 0 \qquad x = 1 \qquad x = -1$$

$$y = -1$$

$$y = y^2 \qquad y^2 - y = 0$$

$$y(y-1) = 0$$

$$y = 0 \qquad y = 1$$

Shell method:

$$x = y^{1/3}$$
$$x = y^2$$

$y = 0$ $\qquad$ $y = 1$

$y = -1$

$$r = y - (-1) = y + 1$$

$$\int_0^1 (y+1)(y^{1/3} - y^2)\,dy.$$